

MAINTENANCE WORK ORDER LOG — PUMP P-104

Period: January 2024 – June 2025 | Unit: Unit-3 | Maintained by: Mechanical Team B

WO #	Date	Type	Action	Technician	Parts Used	Hours
WO-2401	2024-01-15	Preventive	Lubrication — Bearing Housing	Ravi Kumar	Mobil XHP 222 (15g x2)	1.5
WO-2403	2024-02-28	Corrective	Mechanical Seal Replacement	Suresh Babu	John Crane Type 1 Seal	6.0
WO-2407	2024-04-10	Preventive	Lubrication + Vibration Check	Ravi Kumar	Mobil XHP 222 (15g x2)	1.5
WO-2415	2024-06-05	Inspection	Annual Overhaul — Partial	Suresh Babu	Wear rings (2x)	12.0
WO-2422	2024-08-12	Preventive	Lubrication	Ramesh G.	Mobil XHP 222 (15g x2)	1.0
WO-2439	2024-10-20	Corrective	Bearing Replacement (both)	Ravi Kumar	SKF 6205 x2, Grease	8.0
WO-2445	2024-11-08	Corrective	Seal Replacement	Suresh Babu	John Crane Type 1 Seal	5.5
WO-2501	2025-01-22	Preventive	Lubrication	Ravi Kumar	Mobil XHP 222 (15g x2)	1.0
WO-2508	2025-03-12	Corrective	Bearing Replacement — CRITICAL	Ravi Kumar	SKF 6205 x2, Grease 30g	9.0
WO-2514	2025-05-18	Preventive	Lubrication — Overdue	Ramesh G.	Mobil XHP 222 (15g x2)	1.5

Observations

WO-2403 (Feb 2024): Seal failure attributed to dry running during pump restart after emergency shutdown on 2024-02-25. Seal face scored. Replaced with new John Crane Type 1.

WO-2439 (Oct 2024): Vibration reached 8.2 mm/s before shutdown. Lubrication interval had exceeded 600 hours (overdue by 100 hours). Both bearings showed significant wear — grease completely degraded. Root cause: lubrication schedule not followed after WO-2422.

WO-2508 (Mar 2025): Second bearing failure in 5 months. Vibration peaked at 9.1 mm/s. Investigation found lubrication interval again exceeded — last lubrication was WO-2501 (Jan 2025), gap of 51 days / estimated 612 operating hours. RECOMMENDATION: Implement automated lubrication alert at 450-hour threshold.

WO-2514 (May 2025): Lubrication completed. Vibration currently 5.4 mm/s — elevated but within operating limit. Bearing temperature 76°C — monitoring required. Next lubrication due: approximately September 2025 (500 hours from May 18).